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Deep Learning-Based Phishing Email Detection Using LSTM

1. Abstract

Phishing emails are designed to deceive users into revealing sensitive information or performing harmful actions. Manual identification of such emails can be difficult, especially when dealing with a large number of messages. This project presents a deep learning-based phishing email classification system using a Long Short-Term Memory (LSTM) network. The system accepts raw email text and classifies it into two categories: **Safe Email** and **Phishing Email**. A publicly available phishing email dataset was used, and the emails were cleaned, tokenized, converted into numerical sequences, and padded to a fixed length before being given to the LSTM model. The trained model achieved a test accuracy of **98.06%**, demonstrating that LSTM can effectively learn textual patterns associated with phishing emails.

2. Dataset

The dataset used in this project was the **Phishing Email Dataset** obtained from Hugging Face. It initially contained **18,650 emails**, consisting of Safe Email and Phishing Email samples. After removing 16 records with missing email text and 1,097 duplicate emails, the final dataset contained **17,537 unique emails**.

Category	Number of Emails
Safe Email	10,979
Phishing Email	6,558
Total	**17,537**

The labels were converted into numerical values for binary classification:

- **0 → Safe Email**
- **1 → Phishing Email**

3. Data Preprocessing

The email text was processed before training the LSTM model. First, the text was converted to lowercase. HTML tags, URLs, special characters, and unnecessary spaces were removed. The cleaned text was then tokenized using the Keras Tokenizer, with a vocabulary limit of **10,000 words**. Words outside the selected vocabulary were represented using an out-of-vocabulary token.

The resulting text was converted into numerical sequences. Since emails have different lengths, padding and truncation were applied to make every sequence have a fixed length of **500 tokens**. Sequences longer than 500 tokens were truncated, while shorter sequences were padded with zeros.

The dataset was divided using stratified sampling into **80% training, 10% validation, and 10% testing** sets, containing 14,029, 1,754, and 1,754 emails respectively. Stratification was used to maintain a similar Safe/Phishing class distribution across all three sets.

Preprocessing Workflow:

Raw Email → Text Cleaning → Tokenization → Numerical Sequence → Padding/Truncation → LSTM Model

4. Model Architecture

The proposed phishing email classification system uses a sequential **Long Short-Term Memory (LSTM)** neural network. LSTM was selected because email data is sequential text, where the order and context of words can be important for identifying phishing patterns. The model consists of an embedding layer, an LSTM layer, dropout regularization, and fully connected layers.

Architecture

Input Email (500 tokens)
↓
Embedding Layer (128-dimensional vectors)
↓
LSTM Layer (64 units)
↓
Dropout (0.5)
↓
Dense Layer (32 neurons, ReLU)
↓
Output Layer (1 neuron, Sigmoid)
↓
Safe / Phishing

Layer Description

- **Embedding Layer:** Converts each word/token into a 128-dimensional dense vector representation.
- **LSTM Layer:** Processes the sequence of word representations and learns important sequential and contextual patterns from the email.
- **Dropout Layer:** Uses a dropout rate of 0.5 to reduce overfitting during training.
- **Dense Layer:** Contains 32 neurons with ReLU activation to learn higher-level features.
- **Output Layer:** Contains one neuron with sigmoid activation to produce a probability for binary classification.

5. Hyperparameters

Hyperparameter	**Value**
Vocabulary Size	10,000
Maximum Sequence Length	500
Embedding Dimension	128
LSTM Units	64
Dropout Rate	0.5
Dense Layer	32 neurons
Output Activation	Sigmoid
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Batch Size	32
Maximum Epochs	10
Early Stopping Patience	2

The model was trained using the **Adam optimizer** and **binary cross-entropy loss**, which is suitable for binary classification. Early stopping was used by monitoring validation loss and restoring the best model weights to reduce unnecessary training and overfitting.

6. Results and Evaluation

The trained LSTM model was evaluated on the unseen test dataset containing **1,754 emails**. The model achieved a **test accuracy of 98.06%** with a test loss of **0.0789**.

Classification Performance

Class	Precision	Recall	F1-Score
Safe Email	0.99	0.98	0.98
Phishing Email	0.97	0.98	0.97
Overall Accuracy			**98.06%**

The model achieved a **0.97 F1-score for phishing emails** and a **0.98 recall**, indicating that it was able to correctly identify most phishing emails.

Confusion Matrix

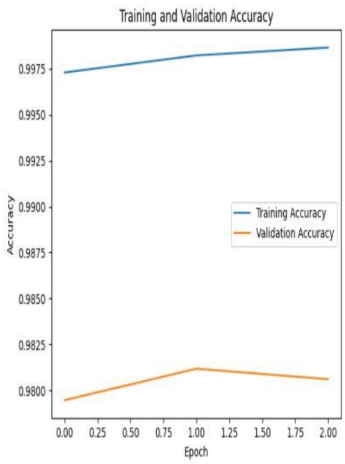
The confusion matrix produced on the test set contained:

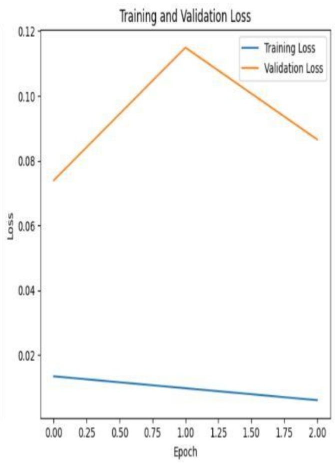
- **1078** correctly classified Safe Emails
- **642** correctly classified Phishing Emails
- **20** Safe Emails incorrectly classified as Phishing
- **14** Phishing Emails incorrectly classified as Safe

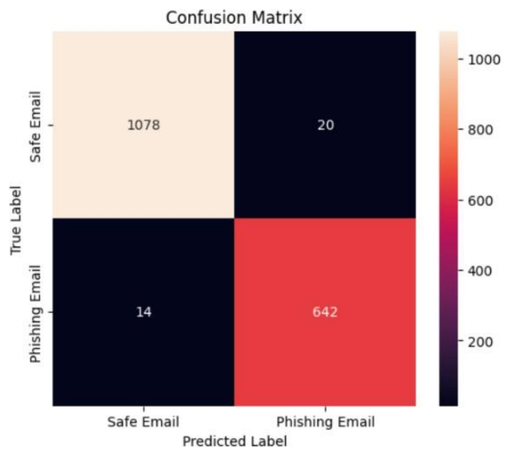
The relatively small number of false negatives indicates that the model successfully detected most phishing emails.

Training Performance

The training and validation accuracy/loss curves were used to monitor model performance across epochs. Training accuracy increased to approximately **99.86%**, while validation accuracy remained around **98%**. The validation loss showed some fluctuation, indicating mild overfitting, but the final test performance remained strong.







7. Conclusion

This project successfully developed a deep learning-based phishing email classification system using an LSTM network. The complete pipeline performs email cleaning, tokenization, padding, feature learning through an embedding layer, sequential processing using LSTM, and binary classification. The model achieved **98.06% test accuracy** and a **0.97 F1-score for phishing emails**. The results demonstrate that LSTM can effectively learn textual patterns useful for distinguishing phishing emails from legitimate emails. The trained system can also accept new raw email text and predict whether it is Safe or Phishing.